

Yiunfan Hu

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Available full-time, on-site Spring 2027 co-op (January–June 2027, 6-month term preferred)

EDUCATION

B.S. Mechanical Engineering | Stony Brook University Stony Brook, NY | Jun 2024–May 2028
GPA 3.97 | University Scholar | Presidential Scholarship (4 yrs) | Dean's List (all semesters) | Senior (U4) standing
Began summer 2024; accelerated junior-level mechanical engineering coursework in 2 years, freeing the entire 3rd year for full-time on-site industry work (Jul 2026 to Jun 2027).

TECHNICAL SKILLS

CAD / Drawing: SolidWorks, Autodesk Inventor, Autodesk Fusion, Onshape, GD&T, DFM/DFA
Manufacturing: CNC milling/turning, manual machining, sheet metal, 3D printing, laser/plasma cutting, welding
Software: C#, C, Python, MATLAB, HTML/CSS/JS, Git

EXPERIENCE

Hardware Development Engineer, Co-op (6-month) | Amazon Robotics North Reading, MA | Jul 2026–present
– Headed the mechanical design of a multipurpose control box for Amazon's robotics platforms, at under 50% of the cost, size, and weight compared to the current design.
– Owned 3 concurrent projects across hardware testing and mechanical design, each from concept through prototype with delivery on or ahead of schedule.
– Worked closely with electrical engineers and the requesting teams; ran design reviews and incorporated feedback on a same-day cycle.

Float Team Lead | Stony Brook Robotics Team – MATE ROV Stony Brook, NY | Sept 2025–Jun 2026
– Led a 3-person team designing a free-floating profiling float for the 2026 MATE ROV International Competition in Newfoundland, Canada.
– Developed a custom 150 psi hydraulic system with 3 independent cylinders and led the main ROV 5-DOF arm design.

Student Researcher | Soft Flyers Group – Reactive Collision UAV Stony Brook, NY | Jun 2025–Jul 2026
– Designed and flight-tested a collision-resistant fixed-wing quadrotor and fabricated major structures including the wings on a CNC router.

Student Researcher Lead | [Shape Memory Alloy \(SMA\) Spring Actuators](#) Stony Brook, NY | May 2024–Jul 2026
– Joint research across 3 labs. Built a test platform for controlled Joule heating of SMA springs, resolving displacement to within 1 mm with a laser rangefinder, temperature to within 2 °C with an IR camera, and force with a load cell.
– Characterized 3 spring configurations; validated published single-spring models and extended them to novel multi-spring networks.

PROJECTS

Team Captain | Combat Robotics, National Havoc Robot League (NHRL) Norwalk, CT | Jun 2023–present
– [12 lb Vertical Spinner \(competed 2023, 2024\)](#): built from scratch in 4 months to a hard competition deadline — CAD, analysis, manufacturing, wiring, driving; iterated revisions between events from analysis of past failure points.
– [12 lb Differential Swerve Drive \(competing 2027\)](#): developed a novel compact drive unit enabling omni-directional movement for small combat robots; currently designing the robot to integrate it.

Solo Maker | ["Javelin" Compact Electric Scooter](#) Stony Brook, NY | Oct 2024–Mar 2025
– Designed and built an ultra-compact folding electric scooter in 6 months on a \$1,000 budget: 26 lb, folds to an extremely portable size [14" × 26" × 4"], 30 mph top speed, 12-mile range, in daily use for commute.
– Achieved minimized folded volume via hinged deck and double-hinged stem design; custom 10s2p Li-ion pack (36 V, 10 Ah) from spot-welded 21700 cells with 260% increase in maximum discharge current over previous OTS battery.

PROFESSIONAL TRAINING

NASA L'SPACE Proposal Writing and Evaluation Academy Virtual | May–Aug 2025
– Developed a technical proposal for a laser ignition system for a rotating detonation rocket engine; reviewed NASA solicitation responses against the official rubric.